

# Enterprise Network Architecture

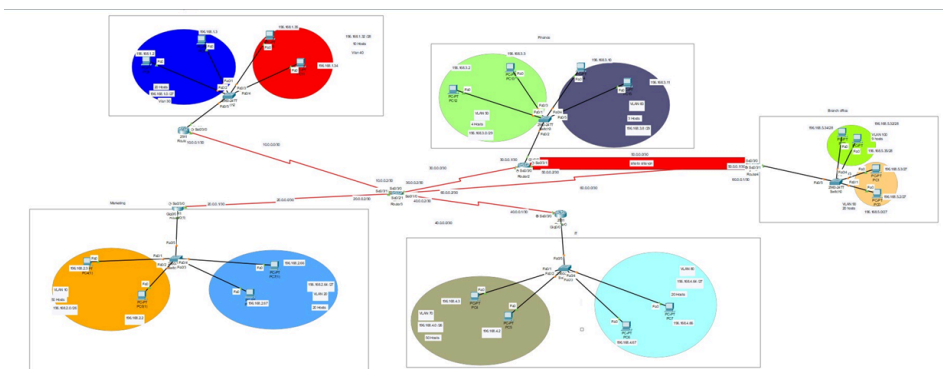
## Using Cisco Packet Tracer.

### PROJECT OVERVIEW

This project presents the design and simulation of a hierarchical enterprise network using Cisco Packet Tracer. The network connects a central site with four departments (Operations, Marketing, Finance, and IT) in addition to a remote Branch office.

Each department is segmented into multiple VLANs to improve performance and security. Inter-VLAN communication is enabled using router-on-a-stick, while OSPF is implemented as the dynamic routing protocol to ensure efficient route distribution across the network. For simplicity, this implementation uses a single OSPF area (Area 0). In a larger enterprise, each department would be configured as a separate area (e.g., Area 10 for Operations, Area 20 for Marketing) to reduce routing table size and improve stability.

A site-to-site IPsec VPN is deployed between the Finance department and the Branch office to provide secure communication for sensitive financial data. The overall design focuses on scalability, logical segmentation, and secure connectivity between network components.



### Network Components

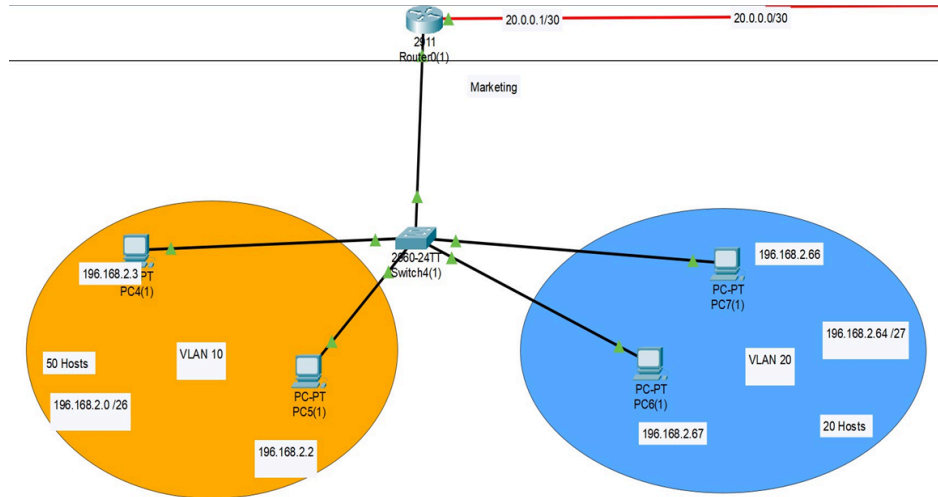
| Device Type | Quantity |
|-------------|----------|
| Routers     | 6        |
| Switches    | 5        |
| PCs         | 16       |

### NETWORK TOPOLOGY

The network follows a hierarchical design consisting of access and core layers. Each department is implemented using a star topology, where end devices connect to an access switch, which in turn connects to a router.

This structure simplifies management and improves fault isolation, as a failure in one segment does not affect the entire network. The core layer (main router) connects all departmental routers and the branch router through WAN serial links, enabling centralized routing and communication across the enterprise.

This modular design allows easy scalability, as new departments or branches can be added without modifying the existing infrastructure.



## IP ADDRESSING SCHEME

VLSM (Variable Length Subnet Masking) was used to efficiently allocate IP address space based on the number of required hosts in each VLAN. This approach minimizes IP wastage compared to fixed-length subnetting.

For example, a network requiring 50 hosts was assigned a /26 subnet, which provides 62 usable IP addresses ( $2^6 - 2$ ), where the two excluded addresses represent the network and broadcast addresses. The gateway simply consumes one of those 62, leaving 61 for end devices. The number of usable IP addresses is calculated as  $2^n - 2$ , where two addresses are reserved for the network and broadcast. The default gateway uses one of the usable addresses.

Each VLAN was assigned a subnet size that closely matches its host requirements. Larger departments were allocated /26 subnets, while smaller networks such as Finance used /29 subnets. WAN links between routers were assigned /30 subnets to support point-to-point communication.

Static IP addressing was configured on all end devices and router interfaces to ensure predictable communication and simplify troubleshooting which is suitable for this scale. For an upscale, a DHCP server may be implemented.

The use of VLSM was not arbitrary. Subnet sizes were carefully selected based on the estimated number of hosts required in each department, taking into account potential future growth. This ensures efficient IP utilization while maintaining scalability without the need for frequent readdressing.

| Department | VLAN     | Network Address | Subnet Mask     | CIDR | Needed Hosts | Total Hosts | Usable Host |
|------------|----------|-----------------|-----------------|------|--------------|-------------|-------------|
| Operations | VLAN 30  | 196.168.1.0     | 255.255.255.224 | /27  | 20           | 32          | 30          |
| Operations | VLAN 40  | 196.168.1.32    | 255.255.255.240 | /28  | 10           | 16          | 14          |
| Marketing  | VLAN 10  | 196.168.2.0     | 255.255.255.192 | /26  | 50           | 64          | 62          |
| Marketing  | VLAN 20  | 196.168.2.64    | 255.255.255.224 | /27  | 20           | 32          | 30          |
| Finance    | VLAN 50  | 196.168.3.0     | 255.255.255.248 | /29  | 4            | 8           | 6           |
| Finance    | VLAN 60  | 196.168.3.8     | 255.255.255.248 | /29  | 3            | 8           | 6           |
| IT         | VLAN 70  | 196.168.4.0     | 255.255.255.192 | /26  | 50           | 64          | 62          |
| IT         | VLAN 80  | 196.168.4.64    | 255.255.255.224 | /27  | 20           | 32          | 30          |
| Branch     | VLAN 90  | 196.168.5.0     | 255.255.255.224 | /27  | 20           | 32          | 30          |
| Branch     | VLAN 100 | 196.168.5.32    | 255.255.255.240 | /28  | 9            | 16          | 14          |

To determine the appropriate subnet size,  $CIDR = 32 - \text{host bits}$ , the number of required *host bits* is calculated using  $\lceil \log_2(\text{needed hosts}) \rceil$ . The total number of addresses is then given by  $2^{(\text{number of host bits})}$ , from which 2 addresses are

reserved for the network and broadcast, leaving the usable host addresses (one is used for gateway, usually `x.x.x.1`) this is why all PCs start with `x.x.x.2`.

## VLAN CONFIGURATION

VLANs were implemented to logically segment the network into separate broadcast domains. This reduces unnecessary traffic and improves overall network performance while enhancing security.

Each department was divided into two VLANs to separate different functional groups. Switch ports were manually assigned to VLANs using access mode, while trunk links were configured between switches and routers to carry multiple VLANs using IEEE 802.1Q encapsulation.

This segmentation ensures that traffic within one VLAN is isolated from others unless explicitly routed.

| VLAN ID | Name    | Department    | Number of Hosts | Subnet          | Switch Port Assignments |
|---------|---------|---------------|-----------------|-----------------|-------------------------|
| 10      | VLAN10  | Marketing     | 50              | 196.168.2.0/26  | Fa0/1, Fa0/2            |
| 20      | VLAN20  | Marketing     | 20              | 196.168.2.64/27 | Fa0/3, Fa0/4            |
| 30      | VLAN30  | Operations    | 20              | 196.168.1.0/27  | Fa0/1, Fa0/2            |
| 40      | VLAN40  | Operations    | 10              | 196.168.1.32/28 | Fa0/3, Fa0/4            |
| 50      | VLAN50  | Finance       | 4               | 196.168.3.0/29  | Fa0/1, Fa0/2            |
| 60      | VLAN60  | Finance       | 3               | 196.168.3.8/29  | Fa0/3, Fa0/4            |
| 70      | VLAN70  | IT            | 50              | 196.168.4.0/26  | Fa0/1, Fa0/2            |
| 80      | VLAN80  | IT            | 20              | 196.168.4.64/27 | Fa0/3, Fa0/4            |
| 90      | VLAN90  | Branch Mgmt   | 20              | 196.168.5.0/27  | Fa0/1, Fa0/2            |
| 100     | VLAN100 | Branch Teller | 9               | 196.168.5.32/28 | Fa0/3, Fa0/4            |

Sample Configuration Commands Run on The IT Dept Switch:

```
enable
configure terminal

vlan 70
name IT_VLAN

vlan 80
name IT_VLAN2

interface fa0/1
switchport mode access
switchport access vlan 70

interface fa0/3
switchport mode access
switchport access vlan 80

interface fa0/5
```

```
switchport mode trunk
```

## INTER-VLAN ROUTING

Each department is connected to an access switch, which in turn connects to a router via trunk links. The switches are responsible for VLAN segmentation, local switching, and forwarding traffic to the router.

Inter-VLAN communication is implemented using a router-on-a-stick configuration. For example, in the Marketing department, each VLAN is assigned a default gateway using router subinterfaces. Switch ports are configured in access mode for end devices and trunk mode for links connecting to the router.

A single physical router interface is divided into multiple subinterfaces (e.g., g0/0.10, g0/0.20), each associated with a specific VLAN allowing efficient communication between VLANs without requiring multiple physical router interfaces, reducing hardware requirements while maintaining flexibility. IEEE 802.1Q encapsulation is used to tag VLAN traffic over the trunk link, allowing the router to correctly route traffic between VLANs.

Sample Configuration Commands Run on IT Router:

```
## IT dept as an example
interface g0/0
no shutdown

interface g0/0.70
encapsulation dot1Q 70
ip address 196.168.4.1 255.255.255.192

interface g0/0.80
encapsulation dot1Q 80
ip address 196.168.4.65 255.255.255.224
```

## ROUTING DESIGN (OSPF)

OSPF (Open Shortest Path First) was implemented as the dynamic routing protocol across the network. It was selected due to its fast convergence, scalability, and suitability for hierarchical network designs.

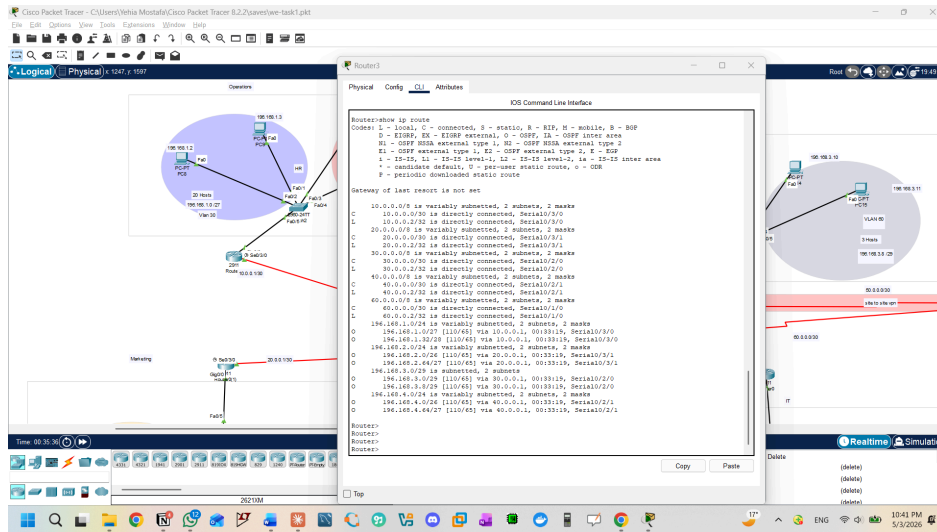
The main router acts as the central OSPF node, while departmental routers advertise their VLAN subnets into the OSPF process. This allows routers to dynamically learn routes and automatically update routing tables in case of network changes.

This eliminates the need for extensive static routing and ensures efficient communication between all departments.

The main router connects to all departmental routers and the branch office through WAN serial interfaces, including:

| Interface   | Network     | Department    |
|-------------|-------------|---------------|
| Serial0/3/0 | 10.0.0.0/30 | Operations    |
| Serial0/3/1 | 20.0.0.0/30 | Marketing     |
| Serial0/2/0 | 30.0.0.0/30 | Finance       |
| Serial0/2/1 | 40.0.0.0/30 | IT            |
| Serial0/1/0 | 60.0.0.0/30 | Branch Office |

### The Main OSPF Routing Table



The routing table of the central router confirms that OSPF is correctly configured across the network. All departmental subnets are learned dynamically (denoted by "O"), each reachable via its respective router interface. This demonstrates successful route propagation and validates the hierarchical routing design.

It is also observed that Branch networks are not present in the OSPF routing table, as they are accessed through a secure IPsec VPN tunnel rather than being advertised via OSPF.

#### Sample Configuration Commands (Main Router and Operations Router):

**## Run On Main Router – advertises only WAN links since it has no local VLANs**

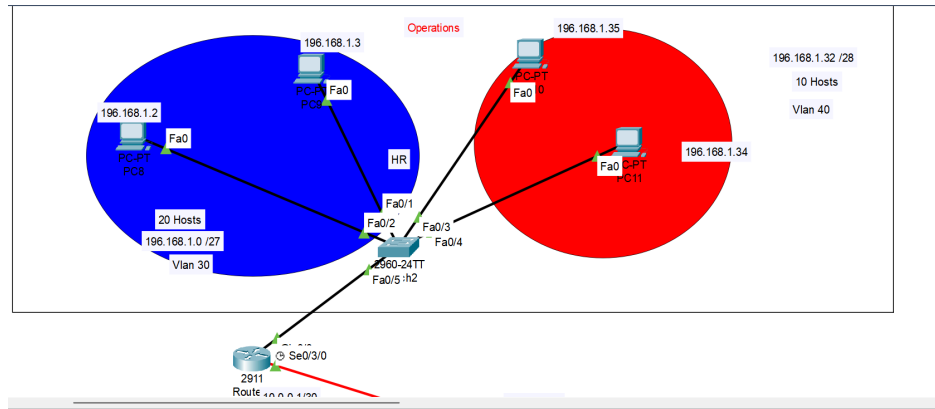
```
router ospf 1
network 10.0.0.0 0.0.0.3 area 0 ## WAN link to Operations
network 20.0.0.0 0.0.0.3 area 0 ## WAN link to Marketing
network 30.0.0.0 0.0.0.3 area 0 ## WAN link to Finance
network 40.0.0.0 0.0.0.3 area 0 ## WAN link to IT
network 60.0.0.0 0.0.0.3 area 0 ## WAN link to Branch
```

**## Run On Departmental Router (Operations as example) – advertises its WAN link**

**## back to the main router AND its two local VLAN subnets so that other**

**## departments can reach Operations hosts through OSPF**

```
router ospf 1
network 10.0.0.0 0.0.0.3 area 0 ## WAN link back to Main Router
network 196.168.1.0 0.0.0.63 area 0 ## VLAN 30 Network ip
network 196.168.1.32 0.0.0.31 area 0 ## VLAN 40 network ip
```



## VPN: SECURITY & BRANCH CONNECTIVITY

A site-to-site IPsec VPN was implemented between the Finance department VLANs (196.168.3.0/29 and 196.168.3.8/29) and the Branch office VLANs (196.168.5.0/27 and 196.168.5.32/28) only to secure sensitive data transmission.

The VPN uses AES encryption for confidentiality and SHA-HMAC for authentication, ensuring data integrity and protection against interception. The tunnel operates using IKE Phase 1 and Phase 2, with a successfully established state (QM\_IDLE). The QM\_IDLE state indicates that IKE Phase 2 has been successfully completed, meaning that the IPsec Security Associations are fully established and ready to securely transmit data.

Only Finance subnets are defined as "interesting traffic" in the VPN configuration. This restricts access to the Branch network from other departments, enforcing security policies and following the principle of least privilege. The VPN endpoints are defined as 50.0.0.2 on the Finance router and 50.0.0.1 on the Branch router. To facilitate this communication, the Finance router has static routes pointing to the Branch LAN subnets through the Branch router IP 50.0.0.1, while the Branch router has static routes pointing to the Finance LAN subnets via the Finance router IP 50.0.0.2. The verification of this is in the verifications section.

### Sample Configuration Commands:

#### On Finance Router2:

```
ip route 196.168.5.0 255.255.255.224 50.0.0.1

crypto isakmp policy 10
 encryption aes
 hash sha
 authentication pre-share
 group 2

crypto isakmp key cisco123 address 50.0.0.1

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
## allowing communication from 2 VLANs to 2 VLANs requires 2^2 combinations.
## as if we say allow X to access A, allow X to access B,
## allow Y to access A, and allow Y to access B
access-list 100 permit ip 196.168.3.0 0.0.0.7 196.168.5.0 0.0.0.31
access-list 100 permit ip 196.168.3.0 0.0.0.7 196.168.5.32 0.0.0.15
access-list 100 permit ip 196.168.3.8 0.0.0.7 196.168.5.0 0.0.0.31
access-list 100 permit ip 196.168.3.8 0.0.0.7 196.168.5.32 0.0.0.15

crypto map VPN-MAP 10 ipsec-isakmp
 set peer 50.0.0.1
```

```
set transform-set VPN-SET
match address 100

interface s0/3/0
crypto map VPN-MAP
```

On Branch Router:

```
ip route 196.168.3.0 255.255.255.248 50.0.0.2

crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2

crypto isakmp key cisco123 address 50.0.0.2

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
## This is a Mirror of Finance Router ACL – source and destination are swapped
access-list 100 permit ip 196.168.5.0 0.0.0.31 196.168.3.0 0.0.0.7
access-list 100 permit ip 196.168.5.32 0.0.0.15 196.168.3.0 0.0.0.7
access-list 100 permit ip 196.168.5.0 0.0.0.31 196.168.3.8 0.0.0.7
access-list 100 permit ip 196.168.5.32 0.0.0.15 196.168.3.8 0.0.0.7

crypto map VPN-MAP 10 ipsec-isakmp
set peer 50.0.0.2
set transform-set VPN-SET
match address 100

interface s0/3/0
crypto map VPN-MAP
```

## SSH

The SSH configuration was implemented on the Operations router. First, the domain name was defined to allow for cryptographic operations, followed by the generation of an RSA key pair used for SSH encryption. The router was then configured to enforce the more secure SSH Version 2 protocol. A local administrative user account was created with an encrypted password to control access. To finalize SSH access, virtual terminal lines were restricted to allow only SSH connections and were set to authenticate using the locally defined credentials. Connectivity to the router was verified using ICMP ping from a remote PC, showing 100% successful packet transmission with stable latency. After confirming network availability, an SSH session was established using the username and password, successfully accessing the router at 10.0.0.1. Here is an example connection and Sample Configuration Commands:

```
hostname OperationsRouter
ip domain-name enterprise.local

crypto key generate rsa
##router prompts for a modulus size
##(e.g., 1024 or 2048 bits),
##determines the strength of the RSA
##key used for encryption.
ip ssh version 2

username admin secret cisco123

line vty 0 4
login local
transport input ssh
```



Instead, a comprehensive verification section is provided to demonstrate that all network components were successfully configured and are functioning as intended, including routing, VLAN segmentation, and secure VPN communication.

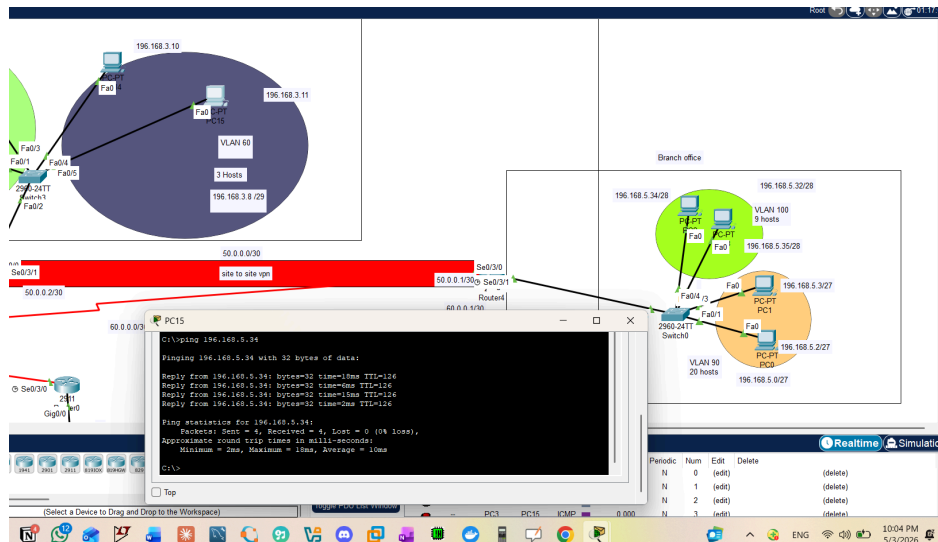
## VPN verification



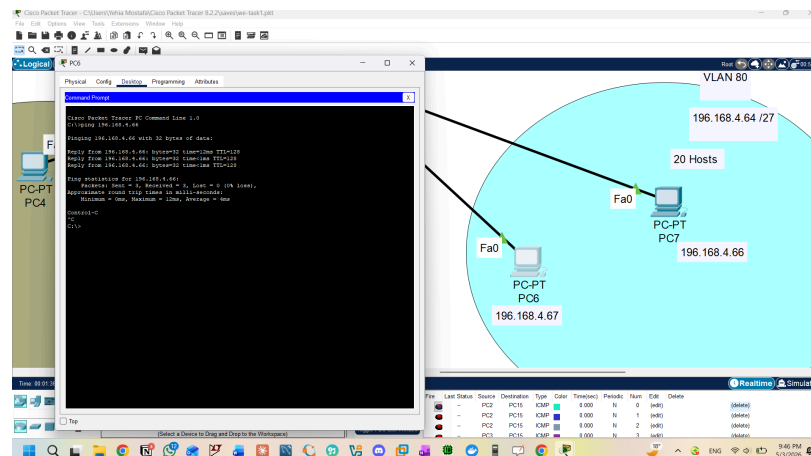
On Finance Router (Router2), the command output shows an active ISAKMP Security Association with the state "QM\_IDLE". Similarly, on Branch Router (Router4), the tunnel is also in the "QM\_IDLE" state.



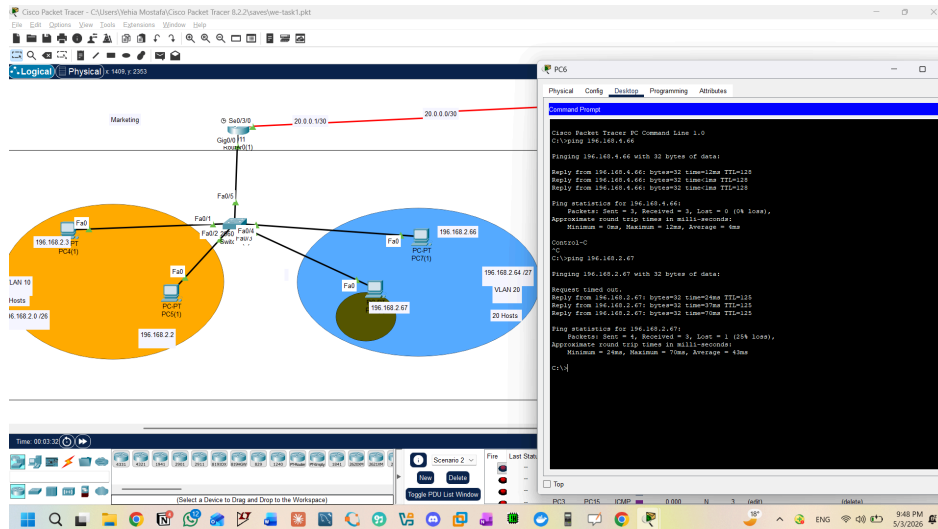
This validates that Layer 2 segmentation is correctly implemented, ensuring isolation between different departments within the network.



## Inter-VLAN communication

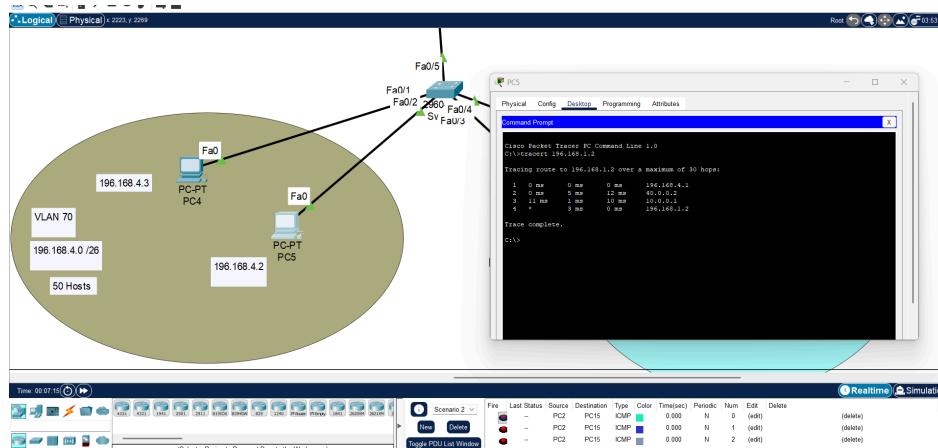


## Department-to-department



Successful ICMP Ping Connectivity between PC6 (196.168.4.67) and PC6(1) (196.168.2.67) shows department-department valid connectivity.

## Traceroute

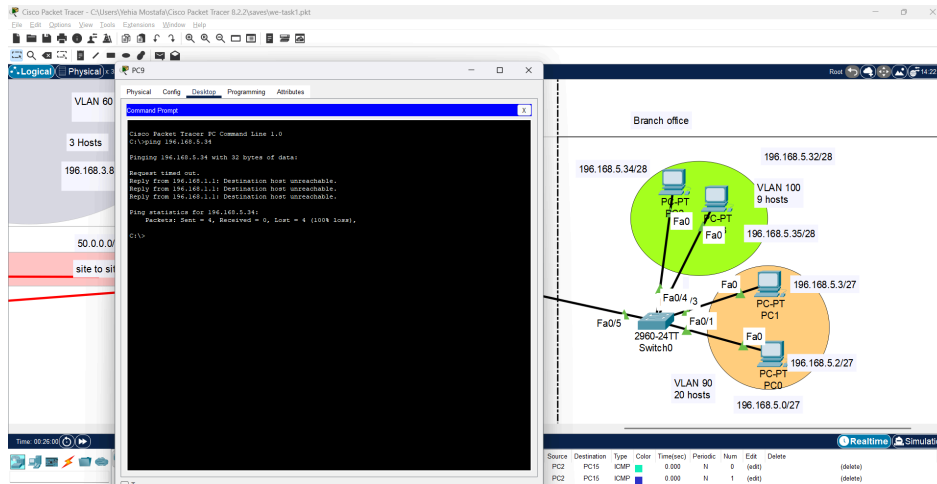


TraceRoute: Branch Router → Main Router → Destination shows correct connectivity through path.

## Restricted access (failure)

Only the Finance department has secure access to the Branch office via the IPsec VPN tunnel. Other departments are intentionally restricted from accessing the Branch network in order to enforce security policies and isolate sensitive financial data.

This selective access control demonstrates how IPsec VPNs can be configured to permit communication only between specific subnets by defining "interesting traffic" using access control lists (ACLs). By modifying these ACL rules, access can be expanded to include additional departments if required.



Connection failure from PC9 (Operations) to the Branch office is expected and valid, as access is restricted to the Finance department through the IPsec VPN tunnel.

## Conclusion

The implementation successfully integrates multiple networking layers, including Layer 2 (VLAN segmentation), Layer 3 (OSPF routing and VLSM addressing), and security mechanisms (IPsec VPN and SSH).